

RIYA MITRA

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Portfolio Website: <https://riya-mitra.github.io/portfolio/>

EDUCATION

UNIVERSITY OF PENNSYLVANIA, Bachelor of Science and Engineering

May 2025

- **Bachelor of Science & Engineering** in Computer Science and Minor in Mechanical Engineering and Engineering Entrepreneurship, GPA: 3.4
- **Relevant Coursework:** AI, Product Design, Scalable Systems & Cloud Computing, HCI, Engineering Entrepreneurship, Negotiations, Economics
- **Organizations:** Society of Women Engineers, Penn Innovation Insider, Wharton Undergraduate Entrepreneurship, Theta Tau Pre-Professional Engineering Fraternity

WORK EXPERIENCE

NVIDIA, Technical Program Manager Intern

May 2024 – August 2024

- Led the development of automation projects through user research to identify inefficiencies in workflows, saving 2-3 hours of daily work and contributing to the successful launch of DRIVE OS 6.0.9.2 and 6.0.9.3 releases, aligning with key product KPIs.
- Spearheaded the development of a custom LLM chatbot powered by AI search, allowing users to easily access critical team information, reducing search time by 50% and enhancing productivity.
- Launched an automated solution for the documentation PM team to streamline the release process from 2 days to 15 minutes, accelerating customer delivery timelines.

MODLAB ROBOTICS, Robotics Software Research Intern

May 2023 – Present

- Led a team of 3 to develop a sustainable digging robot for geothermal cooling, overseeing the design of the linking mechanism, simulation, flow oscillator, and turning mechanism.
- Designed and prototyped an innovative snap-back linking mechanism for a digging robot, achieving a 45% increase in traversal speed and streamlining operational efficiency by 30%.
- Developed a Python simulation for the linking mechanism to dynamically process varying size inputs, optimizing software efficiency and accurately predicting maximum achievable tunnel dimensions.

ORIDEN POWER, Engineering Intern

June 2022 – August 2022

- Led data-driven feasibility studies for 30+ projects to optimize solar design configurations, maximizing energy yield and identified 25 new potential project sites.
- Presented an in-depth research project on new solar technology which increases energy yield by 24% and applied new software models to determine energy production and viability of new technology.
- Engineered a Java application to output the most cost-effective solar panel configurations which increased efficiency and reduced material cost by 30%.

BLOOM, Co-founder, Lead Product Manager

January 2024 – Present

- Spearheaded the development of a gamified household management app, integrating user research to enhance functionality and encouraging collaboration through a virtual flower garden.
- Developed an AI-powered Assistant to automate chore scheduling, predict task completion patterns, and provide personalized reminders based on user behavior.

PROJECTS

GoAero Disaster Management Drone (Senior Design Project) Python, ROS, OpenCV

2024

- Autonomous drone for disaster monitoring, real-time mapping, and payload delivery, utilizing Python and the ODA dataset to develop obstacle detection algorithms in diverse environments.

Insta-lite: Javascript, RAG, Kafka, SQL, React, Java Spark, AWS RDS, ChromaDB

2024

- “Instagram” social media web platform for users to share photos, post status updates, chat with other users, search using an LLM and RAG, and explore real time news from X using Kafka.

SKILLS

Product Management Skills: Road-mapping, User Research, A/B Testing, Wireframing, Figma, Prototyping, Product Lifecycle Management, Customer Journey Mapping, Agile, Scrum

Technical Skills: SQL, Python, Java, Javascript, APIs, AWS, React, Google Cloud Platform, Machine Learning, Jira, Google Analytics, Git